1. Theoretical Background

1.1. Rydberg atoms

1.2. Topological symmetry protected phases

1.2.1. Fock space operators

Fock space

This chapter will start with first recapitalizing the second quantization from a mathematical point of view. Beginning with a Hilbert space $\mathfrak{h}$ for a single particle, this shall be embedded into the large Fock space. The first step is to consider the M-particle space $\otimes_{i=1}^{M}\mathfrak{h}$. For fermions or bosons the space has to be anti-symmetrized or symmetrized respectively. This is accomplished by the action of the (anti-)symmetrizing operator.

$$P_{\pm}^{M} = \frac{1}{M!}\sum_{\sigma\in S_M}(\pm 1)^{\mathrm{sgn}(\sigma)}\sigma$$

S_M is the set of all permutations of the M-particle indices. The Fock space is then the direct sum over the (anti-)symmetrized spaces over all particle numbers

$$\mathfrak{F} = \bigoplus_{M=0}^{\infty} P_{\pm}^{M}\left(\otimes_{i=1}^{M}\mathfrak{h}\right) =: \bigoplus_{M=0}^{\infty}\mathfrak{f}_M$$

A single particle operator $A_s : \mathfrak{h} \to \mathfrak{h}$ can then be embedded into the Fock space by lifting it into the M-particle spaces via

$$A^{(M)} = \sum_{i=1}^{M}\bigotimes_{j=1}^{M}\left(\delta_{i\neq j}\,\mathbb{1} + \delta_{i=j}\,A_s\right)$$

The operator on the Fock space A then corresponds to the sum over all lifted operators

$$A = \bigoplus_{M=0}^{\infty} A^{(M)}$$

As can already be seen from this expression operators, defined like this, conserve the particle number, as they are block diagonal with respect to the constant particle number blocks. In order to introduce operators which can mix particle numbers one has to already start with operators on the multi-particle space

$$B_{K,L} : \otimes_{i=1}^{K}\mathfrak{h} \to \otimes_{i=1}^{L}\mathfrak{h},$$

whereas the operator $B_{K,L}$ has to conserve the symmetry of states. One then can lifts this operator into higher particle space as an operator $B^{(M)} : \mathfrak{f}_M \to \mathfrak{f}_{M+\Delta}$, where $(\Delta = L - K)$, by considering all $M!\,(M+\Delta)!/(K!\,L!)$ possible connections, between the spaces. The full operator is then the sum

$$B = \bigoplus_{M=\max(K,K-\Delta)}^{\infty} B^{(M)}$$

Second quantization

For an orthonormal subset $(\alpha_1,\ldots,\alpha_N) \subseteq B$ of a basis of the one-particle space $\mathfrak{h}$ with prevalence-vector $(n_1,\ldots,n_N)$, $\sum_{i=1}^{N} n_i = M$, one can construct a family $\Phi = (\phi_1,\ldots,\phi_M)$ of M states by stacking the states together according to their prevalence. From this family an (anti-)symmetric state $|\phi\rangle$ can be created via

$$
\begin{aligned}
|\Phi\rangle &= \frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,P_\pm^M\left(\Phi\right)\\
&= \frac{1}{\sqrt{M!\,n_1!\cdots n_N!}}\sum_{\sigma\in S_M}\phi_{\sigma(1)}\otimes\cdots\otimes\phi_{\sigma(M)}\\
&=: |n_1\phi_1,\dots,n_N\phi_N\rangle
\end{aligned}
\tag{1.1}
$$

One can then introduce creation and annihilation operators for a one-particle state $\psi\in B$ acting on any M-particle state by defining

$$
\begin{aligned}
a^\dagger(\psi)\;|\Phi\rangle &= \sqrt{M+1}\,\frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,P_\pm^{M+1}\left(\psi\otimes\phi_1\otimes\cdots\otimes\phi_M\right)\\
&= \sqrt{n_\psi+1}\;|n_1\phi_1,\dots,(n_\psi+1)\psi,\dots,n_N\phi_N\rangle\\
a(\psi)\;|\Phi\rangle &= \sqrt{M}\;\langle\psi|\Phi\rangle_{\text{partial}} = \sqrt{M}\;\frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,n_\psi\,\frac{(M-1)!}{M!}\,P_\pm^{M-1}\left(\phi_1\otimes\cdots\otimes\phi_{M-1}\right)\\
&= \frac{\sqrt{(M-1)!}}{\sqrt{n_1!\cdots(n_\psi-1)!\cdots n_N!}}\,\sqrt{n_\psi}\,P_\pm^{M-1}\left(\phi_1\otimes\cdots\otimes\phi_{M-1}\right)\\
&= \sqrt{n_\psi}\;|n_1\phi_1,\dots,(n_\psi-1)\psi,\dots,n_N\phi_N\rangle
\end{aligned}
$$

where for a vector $|\alpha\rangle = \alpha_1\otimes\cdots\otimes\alpha_M$ the above scalar product is defined as

$$
\langle\psi|\alpha\rangle_{\text{partial}} = \langle\psi|\alpha_1\rangle\;\alpha_2\otimes\cdots\otimes\alpha_M.
$$

The operation $(M+1)!\,P_\pm^{M+1}\left(\psi\otimes\phi_1\otimes\cdots\otimes\phi_M\right)$ is equivalent to inserting the vector ψ at every position of every tensor product, which is involved in building the state $|\phi\rangle$. In the next step I want to present, how one can represent the multi-particle operators in terms of creation and annihilation operators. Therefore one resorts to the basis of the one-particle space $\mathfrak{h}$ and define $a_i := a(\phi_i)$. Let us consider a generic operator $A_s = |\phi_\alpha\rangle\langle\phi_\beta|$ written in bra-ket notation in the single-body space. The lifted operator $A^{(M)}$ on in the M-particle space takes the form

$$
A^{(M)} = \sum_{i=1}^{M}\mathbb{1}\otimes\cdots\otimes|\phi_\alpha\rangle\langle\phi_\beta|\otimes\cdots\otimes\mathbb{1}
$$

Its action onto a M-particle state has the effect that it takes the partial scalar product of the state with ϕ_β at all positions, which are, however, equivalent, and correspondingly inserts then at every position of the reduced state, i.e. the now empty positions, the vector ϕ_α. In other words the action of $A^{(M)}$ is such, that for every summand in Eq. (1.1) it replaces at every possible position once the vector ϕ_β by ϕ_α. If one defines the family Φ', where one ϕ_β in Φ is replaced by a ϕ_α, the action can therefore be written as

$$
\begin{aligned}
A^{(M)}\,|\Phi\rangle &\to \frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,\frac{n_\beta}{M}\,P_\pm^{M-1}\left(\Phi/\{1\cdot\phi_\beta\}\right) \to \frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,\frac{n_\beta}{M}\,\frac{M!}{(M-1)!}\,P_\pm^M\left(\Phi'\right)\\
&= \frac{\sqrt{M!}}{\sqrt{n_1!\cdots n_N!}}\,n_\beta\,P_\pm^M\left(\Phi'\right) = \sqrt{n_\beta\left(n_\alpha+1\right)}\;|\Phi'\rangle\\
&= a_\alpha^\dagger a_\beta\;|\Phi\rangle
\end{aligned}
$$

In this way one can construct for a generic operator A_s the creation and annihilation operator representation of the lifted Fock space operator as

$$
A = \sum_{i,j=1}^{\dim(\mathfrak{h})}\langle e_i|A_s|e_j\rangle\;a_i^\dagger a_j
\tag{1.2}
$$

This is similar to a basis representation for a single body operator, while avoiding to explicitly write the symmetrization overhead.

For the other type of Fock space operators originating from multi-particle operators, one proceeds in the same way. For a function B from the previous section, one can express

$$B = \sum_{i_1=1}^{\dim(\mathfrak{h})} \sum_{i_2=1}^{i_1} \cdots \sum_{i_L}^{i_{L-1}} \sum_{j_1=1}^{\dim(\mathfrak{h})} \sum_{j_2=1}^{j_1} \cdots \sum_{j_K=1}^{j_{K-1}} \langle l_{i_1} \otimes \cdots \otimes l_{i_L} | B_{K,L} | k_{j_1} \otimes \cdots \otimes k_{j_K} \rangle \, a_{j_1}^\dagger \cdots a_{i_L}^\dagger a_{j_1} \cdots a_{j_K}$$

For an operator $B_{2,0}$ with $K = 2$, $L = 0$ this takes the explicit form

$$B = \sum_{j_1=1}^{\dim(\mathfrak{h})} \sum_{j_2=1}^{j_1} B_{2,0} \, |k_{j_1} \otimes k_{i_2}\rangle \, a_{i_1} \, a_{i_2} \tag{1.3}$$

In order to validate this for the above example, take the lifted operator D_2 with

$$D_2 = \sum_{\alpha,\beta=1}^{\dim(\mathfrak{h})} \langle \phi_\alpha | \langle \phi_\beta |$$
$$D^{(M)} = \frac{1}{\sqrt{M(M-1)}} \sum_{\alpha,\beta=1}^{\dim(\mathfrak{h})} \sum_{i,j=1;\, i\neq j}^{M} \bigotimes_{k=1}^{M} \left[\mathbb{1} + \delta_{k,i} \left(\langle \phi_\alpha | - \mathbb{1} \right) + \delta_{k,j} \left(\langle \phi_\beta | - \mathbb{1} \right) \right]$$

where the factor $1/\sqrt{M(M-1)}$ is necessary to achieve a well defined output state. When acting with $D^{(M)}$ on a state, one can exploit the symmetry property of the state in order simplify the calculation

$$\begin{aligned}
D^{(M)} |\Phi\rangle &= \sqrt{M(M-1)} \sum_{\alpha,\beta=1}^{\dim(\mathfrak{h})} \langle \phi_\alpha | \otimes \langle \phi_\beta | \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} |\Phi\rangle \\
&= \sum_{\alpha,\beta=1}^{\dim(\mathfrak{h})} \sqrt{M-1} \, \langle \phi_\alpha | \Phi' \rangle_{\text{partial}} \qquad \text{with } |\Phi'\rangle = \sqrt{M} \, \langle \phi_\beta | \Phi \rangle_{\text{partial}} \\
&= \sum_{\alpha,\beta=1}^{\dim(\mathfrak{h})} a_\alpha \, a_\beta \, |\Phi\rangle
\end{aligned}$$

Both Eq. (1.2) and Eq. (1.3) can be written in matrix form after introducing the vector $\mathbf{a} = (a_1, \ldots, a_{\dim(\mathfrak{h})})$ and defining appropriate matrices $\mathbf{A}$ and $\mathbf{B}$.

$$A = (\mathbf{a}^\dagger)^t \mathbf{A} \, \mathbf{a} \qquad B = \mathbf{a}^t \mathbf{B} \, \mathbf{a}$$

For the bosonic case these matrices take the form

$$\mathbf{A} = \left(\langle \phi_i | A_s | \phi_j \rangle \right)_{i,j} \qquad \mathbf{B} = \left(B_{2,0} \, |\phi_i \otimes \phi_j \rangle \right)_{i,j}$$

1.2.2. Introduction to topological phases

Time-reversal symmetry

Time reversal symmetry is represented by an anti-unitary operator T. The effect of time-reversal on an operator is the following.

$$\hat{O} \to T \hat{O} T^{-1},$$

with $T^{-1} = T^\dagger$, as the norm of vectors has to be conserved. T can be decomposed into two operators $T = \tau K$, where K is the complex conjugation operator acting on everything to the right. T can satisfy one of conditions

$$T^2 = \pm 1$$

For spin-less particles in real space usually $\tau = \text{id}$ and $T^2 = 1$, whereas for particles with spin $\tau = -i\sigma_y$ can be chosen in real space, which fulfills $T^2 = -1$.
Take a look at the momentum space Hamiltonian without spin degrees of freedom

$$H = \sum_{\mathbf{k}} H(\mathbf{k}) \, a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$
$$\text{with} \quad a_{\mathbf{k}} = \frac{1}{\sqrt{N^d}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\mathbf{R}} \, a_{\mathbf{R}}$$

Time-reversal transforms the operators according to

$$\begin{aligned} T a_{\mathbf{R}} T^{-1} &= a_{\mathbf{R}} \\ \Rightarrow \quad T a_{\mathbf{k}} T^{-1} &= a_{-\mathbf{k}} \end{aligned}$$

Symmetry regarding time-reversal ($H = THT^{-1}$) has thus the form

$$\begin{aligned} H &= THT^{-1} \\ &= \sum_{\mathbf{k}} H^*(\mathbf{k})\, a^\dagger_{-\mathbf{k}} a_{-\mathbf{k}} \\ &= \sum_{\mathbf{k}} H^*(-\mathbf{k})\, a^\dagger_{\mathbf{k}} a_{\mathbf{k}} \\ \Longleftrightarrow \quad H(\mathbf{k}) &= H^*(-\mathbf{k}) \end{aligned}$$

Particle-hole symmetry

Chiral symmetry

Bulk-boundary correspondence

The bulk-boundary correspondence states a connection between topological indices, which can be calculated for Hamiltonians defined on an infinite d-dimensional lattice, with properties on the edge of a half-space (half-lattice) Hamiltonian. The Hamiltonian of the full space H, in the following also called bulk Hamiltonian, is an operator on a single-particle Hilbert space

$$\begin{aligned} \mathfrak{h} &= \mathbb{C}^N \otimes \ell^2(\mathbb{Z}^d) \\ H &: \mathfrak{h} \to \mathfrak{h}, \end{aligned}$$

where $\mathbb{Z}^d$ accounts for the lattice and the internal degrees of freedom are represented in $\mathbb{C}^N$. The half-space is simply defined by

$$\tilde{\mathfrak{h}} = \mathbb{C}^N \otimes \ell^2(\mathbb{Z}^{d-1} \times \mathbb{N})$$

and one defines the corresponding half-space Hamiltonian $\hat{H} : \tilde{\mathfrak{h}} \to \tilde{\mathfrak{h}}$. The bulk-boundary correspondence can come in different mathematical forms [1]. One particular version relates the Chern number calculated for a band gap of the bulk Hamiltonian to a charge current in the half-space. For two dimensions this correspondence reads [1, 2]

$$-\frac{1}{2\pi} \hat{\mathcal{T}}\left(g(\hat{H}_{k_1})\, \nabla_{k_1} \hat{H}_{k_1} \right) = \mathrm{Ch}_2(P_F)$$

Let us disassemble the relation. $\mathrm{Ch}_2(P_F)$ is the Chern number computed for the Fermi projection, which is the projection on the energy bands of the bulk Hamiltonian below the Fermi energy μ_F, whereas the Fermi energy is located in the bulk gap. The chern number is an index which is computed exclusively from the bulk Hamiltonian and takes integer values. The left hand-side of the equation, however, is a property of the boundary Hamiltonian an represents a charge current. $\mathcal{T}$ is the trace per unit volume and the hats denote boundary operators. $\hat{H}_{k_1}$ is the half-space Hamiltonian transformed into momentum space along the periodic first dimension, whereas g is an arbitrary continuous function with support exclusively in the bulk energy gap.

Therefore $g(\hat{H}_{k_1})$ is only non-vanishing for any g of this property, if the boundary Hamiltonian has eigenenergies inside the bulk energy gap. Thus for non vanishing bulk chern number, the half-space system must possess energies inside the bulk gap.

On the other hand, if the Fermi level lies inside of a bulk energy gap, one can prove the following property [1] (Proposition 2.4.10)

$$\left| \langle x, n | g(\hat{H}) | y, m \rangle \right| \le A_M (1 + |x - y|)^{-M} e^{-\beta(n+m)}, \quad n, m \in \mathbb{N},\ x, y \in \mathbb{Z}^{d-1}$$

here M can be any integer and A_M, β are strictly positive constants. n, m are the positions in the dimension which was cut to half, when introducing the half-space system. From this follows that an eigenstate of $\hat{H}$ with energy inside the bulk gap is localized at the boundary. This can be understood when considering the amplitude of such an eigenstate $|\psi\rangle$ at position (x,n). One can exploit that $g(\hat{H})\,|\psi\rangle = \alpha\,|\psi\rangle$ with $\alpha \neq 0$ for appropriate g.

$$
\begin{aligned}
|\langle x,n|\psi\rangle| &= \frac{1}{|\alpha|}\,\left|\langle x,n|g(\hat{H})|\psi\rangle\right| = \frac{1}{|\alpha|}\left|\sum_{y\in\mathbb{Z}^{d-1},m\in\mathbb{N}} \langle x,n|g(\hat{H})|y,m\rangle\,\langle y,m|\psi\rangle\right| \\
&\leq \frac{1}{|\alpha|}\sum_{y\in\mathbb{Z}^{d-1},m\in\mathbb{N}} |\,\langle x,n|g(\hat{H})|y,m\rangle\,|\,|\,\langle y,m|\psi\rangle\,| \\
&\leq \frac{1}{|\alpha|}\sum_{y\in\mathbb{Z}^{d-1},m\in\mathbb{N}} A_M(1+|x-y|)^{-M}e^{-\beta(n+m)}||\psi|| \\
&= \frac{||\psi||}{|\alpha|\,(1-e^{-\beta})}\sum_{y\in\mathbb{Z}^{d-1}} A_M(1+|x-y|)^{-M}e^{-\beta\,n}
\end{aligned}
$$

If $M \geq d$ is chosen the sum over y converges and one can infer the exponential decay of the eigenstate with distance n from the boundary.

$$
|\langle x,n|\psi\rangle| \leq C_M\,e^{-\beta\,n}
$$

This behavior can be made intuitively plausible, if one takes a look at the explicit form of the half-space Hamiltonian

$$
\hat{H} = \Pi_d H \Pi_d^\dagger + \tilde{H}.
$$

Here Π_d projects the bulk Hamiltonian on the half-space and $\tilde{H}$ constitutes a correction due to the rise of a boundary, which acts only on a finite range from this edge. Thus, if an eigenstate of the half-space Hamiltonian has significant support away from the boundary inside the half-space bulk, the eigenstate is mainly determined by the bulk-contribution, due to the relative size of bulk and boundary. In consequence the eigenstate is close to an bulk eigenstate of the whole space, where no energies inside the bulk gap are found. The only way that an eigenenergy inside the gap can occur, is that the corresponding eigenstate is mainly localized at the boundary and is mainly influenced by the boundary correction $\tilde{H}$.

In summary these two properties of eigenenergies of the half-space Hamiltonian inside the bulk gap with eigenstates exponentially localized at the boundary are the main characteristics of topological protected non-trivial phases.

1.2.3. Bogoliubov-de Gennes Hamiltonian

As stated in Section 1.2.2 about the bulk boundary correspondence the topological properties and there analysis is restricted to single-particle Hamiltonians. A generic Hamiltonian that describes many physical systems and is also relevant for the present thesis is the quadratic Hamiltonian, written in general form as

$$
\mathcal{H} = \frac{1}{2}(\mathbf{a}^\dagger)^t h\,\mathbf{a} + \frac{1}{2}\mathbf{a}^t\overline{h}\,\mathbf{a}^\dagger + \frac{1}{2}(\mathbf{a}^\dagger)^t\Delta\,\mathbf{a}^\dagger + \frac{1}{2}\mathbf{a}^t\Delta^\dagger\,\mathbf{a}, \qquad h = h^\dagger, \quad \Delta = \Delta^t \tag{1.4}
$$

It can be written in matrix form as

$$
\mathcal{H} = \frac{1}{2}\begin{pmatrix}(\mathbf{a}^\dagger)^t & \mathbf{a}^t\end{pmatrix}\tilde{A}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^\dagger\end{pmatrix}, \qquad \tilde{A} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & \overline{h}\end{pmatrix}
$$

This Hamiltonian is, however, not of the single-particle form due to the coupling terms proportional to Δ, which are two-particle operators. This problem can be circumvented by interpreting the creation part of the stacked operator vector as operators in a hole-space, i.e. hole-annihilation operators and then treating holes and particles as independent degrees of freedom.

To show how this reinterpretation can be carried out, take for example the last term of the Hamiltonian. It takes two particles and annihilates them. According to Section 1.2.1 it can be viewed as originating from a bilinear form on the two particle space, reducing the particle number by two

$$
\boldsymbol{\Delta}^\dagger : \mathfrak{h}\times\mathfrak{h} \to \mathbb{C} : |\psi\rangle\,|\phi\rangle \mapsto \boldsymbol{\Delta}^\dagger\,(|\psi\rangle\,|\phi\rangle)
$$

Plugging in a certain vector $|\psi\rangle$ and leaving the other spot open, we can, however, define a map $\tilde{\mathbf{\Delta}}^\dagger(|\psi\rangle) : \mathfrak{h} \to \mathcal{L}(\mathfrak{h}, \mathbb{C}) \to \mathfrak{h}^*$, where $\mathcal{L}(\mathfrak{h}, \mathbb{C})$ is the space of linear maps from $\mathfrak{h}$ to $\mathbb{C}$. But each such map can be represented in the dual space by a vector $\mathcal{L}(\mathfrak{h}, \mathbb{C}) \to \mathfrak{h}^* : \mathbf{\Delta}^\dagger(|\psi\rangle, \cdot) \mapsto |\tilde{\Delta}^\dagger(\psi)\rangle \in \mathfrak{h}^*$ such that $\mathbf{\Delta}^\dagger(|\psi\rangle|\phi\rangle) = \langle\tilde{\Delta}^\dagger(\psi)|\phi\rangle$. The linearity of $\mathbf{\Delta}^\dagger$ results in

$$|\tilde{\Delta}^\dagger(\lambda\psi)\rangle = \bar{\lambda}\,|\tilde{\Delta}^\dagger(\psi)\rangle$$

This makes $\tilde{\mathbf{\Delta}}^\dagger$ an anti-linear operation, instead of the required linear one. This issue can be resolved by introduce a different space $\overline{\mathfrak{h}}$ which possesses a scalar multiplication of the form

$$\cdot : \mathbb{C} \times \overline{\mathfrak{h}} \to \overline{\mathfrak{h}} : (\lambda, |\varphi\rangle) \mapsto \bar{\lambda}\,|\varphi\rangle$$

By this way one can define a linear operator over single particle Hamiltonians

$$\mathbf{\Delta}_s^\dagger : \mathfrak{h} \to \overline{\mathfrak{h}} : |\psi\rangle \mapsto |\tilde{\Delta}^\dagger(\psi)\rangle$$

In a similar way one can define

$$\mathbf{\Delta} : \mathbb{C} \to \mathfrak{h} \times \mathfrak{h} : 1 \mapsto \Delta\,|\psi\rangle\,|\phi\rangle$$
$$\rightsquigarrow \mathbf{\Delta}_s : \overline{\mathfrak{h}} \to \mathfrak{h} : |\psi\rangle \mapsto \langle\psi|\ \mathbf{\Delta}(1) = \Delta\,|\phi\rangle$$

where $\mathbf{\Delta}_s$ can be divided into a simple linear map $M : \overline{\mathfrak{h}} \to \mathfrak{h}$ that maps each basis vector $|b\rangle_{\overline{\mathfrak{h}}}$ in $\overline{\mathfrak{h}}$ to the appropriate basis vector $|b\rangle_{\mathfrak{h}}$ in $\mathfrak{h}$, thus $\bar{\lambda}\,|b\rangle_{\overline{\mathfrak{h}}} \mapsto \lambda\,|b\rangle_{\mathfrak{h}}$, followed by adjunction and multiplication with $\mathbf{\Delta}(1)$.

$$\mathbf{\Delta}_s : \overline{\mathfrak{h}} \to \mathfrak{h} : |\psi\rangle \mapsto (M(|\psi\rangle))^\dagger\ \mathbf{\Delta}(1) = \Delta\,|\phi\rangle$$
$$\rightsquigarrow \mathbf{\Delta} : \mathbb{C} \to \mathfrak{h} \times \mathfrak{h} : 1 \mapsto M\left(|\psi\rangle_{\overline{\mathfrak{h}}}\right)\mathbf{\Delta}_s(|\psi\rangle_{\overline{\mathfrak{h}}})$$

On the other hand the operators proportional to h are already of the desired form and can be seen as originating from a single-particle operator. Therefore one interprets the first term of Eq. (1.4) as acting on the particle space and originating from

$$\mathbf{h} : \mathfrak{h} \to \mathfrak{h}$$

and the second term as acting on the hole space and originating from

$$\overline{\mathbf{h}} : \overline{\mathfrak{h}} \to \overline{\mathfrak{h}}$$

At this state one identify the space $\mathfrak{h}$ with the single-particle space and $\overline{\mathfrak{h}}$ with the single-hole space, where particle and holes are interpreted as independent degrees of freedom. The total space is than represented by the Nambu-spinor $(|\psi\rangle_{\mathfrak{h}}, |\varphi\rangle_{\overline{\mathfrak{h}}})$ with particle degrees of freedom $|\psi\rangle$ and hole degrees of freedom $|\phi\rangle$. This lets the effect of such an single-particle Hamiltonian on a Nambu-state be written as

$$\mathcal{H}_s \begin{pmatrix} |\psi\rangle \\ |\phi\rangle \end{pmatrix} = \begin{pmatrix} \mathbf{h} & \mathbf{\Delta} \\ \mathbf{\Delta}^\dagger & \overline{\mathbf{h}} \end{pmatrix} \begin{pmatrix} |\psi\rangle \\ |\phi\rangle \end{pmatrix}$$

By defining a basis B of $\mathfrak{h}$ and C of $\overline{\mathfrak{h}}$, one can construct a basis of the Nambu space $D = B \oplus C$. In this way one can lift the just defined single-particle Hamiltonian into Fock space via second quantization. By defining the matrices

$$h' = \Big(\langle b_i|\mathbf{h}|b_j\rangle\Big)_{i,j} = h$$
$$\overline{h}' = \Big(\langle c_i|\overline{\mathbf{h}}|c_j\rangle\Big)_{i,j} = \overline{h}$$
$$\Delta' = \Big(\langle b_i|\mathbf{\Delta}|c_j\rangle\Big)_{i,j} = \Delta$$
$$(\Delta^\dagger)' = \Big(\langle c_i|\mathbf{\Delta}^\dagger|b_j\rangle\Big)_{i,j} = \Delta^\dagger$$

By then defining the vector of annihilation operators $\alpha = (a(b_1), \dots, a(b_L))$ and $\beta = (a(c_1), \dots, a(c_L))$, $L = \dim(\mathfrak{h})$, one can write the quadratic Hamilonian in particle-hole form

$$\mathcal{H} = (\alpha^\dagger)^t\, h\, \alpha + (\beta^\dagger)^t\, \overline{h}\, \beta + (\beta^\dagger)^t\, \Delta^\dagger\, \alpha + (\alpha^\dagger)^t\, \Delta\, \beta$$

Momentum space representation

The space $\mathfrak{h} = \mathbb{C}^N \otimes \ell^2(\mathbb{Z}^d)$ can be structured by the canonical basis for $\mathbb{C}^N$, denoted as G in the following, combined with a basis for $\ell^2(\mathbb{Z}^d)$. As basis of $\ell^2(\mathbb{Z}^d)$ one can choose all functions $\{f_z|z \in \mathbb{Z}^d\}$ with $f_z : \mathbb{Z}^d \to \mathbb{C} : x \mapsto \delta_{x,z}$, which is the canonical discrete position basis $f_z =: |z\rangle$. For $\overline{\mathfrak{h}}$ one can use the same basis for the postion combined with a basis of $\mathbb{C}^N$ with a different scalar multiplication. Each single particle operator can be separated into operators that act on the internal system $(\mathbb{C}^N)$ of a unit cell and one, that mediates between unit cells u^x. In a generic form one can write an opertor with range $\mathcal{R}$ in the following form. The range is here the set of distances x with non-zero coupling.

$$\mathbf{h} = \sum_{x \in \mathcal{R}} h_x u^x$$

where $h_x : \mathbb{C}^N \to \mathbb{C}^N$ treats the internal degrees of freedom and $u^x : \ell^2(\mathbb{Z}^d) \to \ell^2(\mathbb{Z}^d) : |z\rangle \mapsto |z - x\rangle$ is just the shift operator. The Fourier transform is a map from $\mathfrak{h}$ to the d-dimensional torus $\mathcal{F} : \ell^2(\mathbb{Z}^d) \to \mathbb{T}^d = (-\pi, \pi] \times \cdots \times (-\pi, \pi]$. Fourier transforming a state in the braket notation can be written as an operator $\int_{\mathbb{T}^d}^{\oplus} dk \sum_x e^{-ikx} |k\rangle \langle x|$. With respect to k the sum goes over a continuous parameter space which is indicated by the integral symbol. $\{|k\rangle \, |k \in \mathbb{T}^d\}$ is here a basis of the d-dimensional torus in the same way as $\{|x\rangle \, |x \in \mathbb{Z}^d\}$ is a basis for the discrete lattice. By splitting a state in internal and lattice degrees of freedom one can write $|\psi\rangle = |\psi_{\text{int}}\rangle |\psi_{\text{Lat}}\rangle \in \mathfrak{h}$, $|\phi\rangle = |\phi_{\text{int}}\rangle |\phi_{\text{Lat}}\rangle \in \overline{\mathfrak{h}}$, and as usual $\psi(y) = |\psi_{\text{int}}\rangle \langle y|\psi_{\text{Lat}}\rangle$, where y is the position of a unit cell.

$$\begin{aligned} \mathcal{F}(\psi)(k) &= \langle k|\mathcal{F}(\psi)\rangle = \sum_y e^{-iky}\psi(y) && \text{particle} \\ \mathcal{F}(\phi)(k) &= \sum_y e^{iky}\phi(y) && \text{hole} \end{aligned}$$

A new basis for $|\psi\rangle_k$ is therefore $G_k := \{|g_{\text{int}}\rangle |k\rangle \, |g \in G, \; k \in \mathbb{T}^d\}$. One can determine the Fourier transform of the operators by their action on the states.

$$\begin{aligned} \mathcal{F}(\mathbf{h}\psi)(k) &= \sum_{y,x} e^{-iky} \, h_x u^x \psi(y) \\ &= \sum_{y,x} e^{-iky} \, h_x \psi(y - x) \\ &= \sum_{z,x} e^{-ikz} e^{-ikx} \, h_x \psi(z) \\ &= \sum_x e^{-ikx} \, h_x \mathcal{F}(\psi)(k) \\ &=: h(k) \, \mathcal{F}(\psi)(k) =: h(k) \, \psi(k) \end{aligned}$$

As $\overline{h} = \sum_x \overline{h}_x u^x$, where the bar means complex conjugation, we can write

$$\begin{aligned} \mathcal{F}(\overline{\mathbf{h}}\phi)(k) &= \sum_{y,x} e^{iky} \, \overline{h}_x u^x \phi(y) \\ &= \sum_x e^{ikx} \, \overline{h}_x \mathcal{F}(\phi)(k) = \overline{\sum_x e^{-ikx} \, h_x} \mathcal{F}(\phi)(k) \\ &= \overline{h(k)} \, \phi(k) \end{aligned}$$

For $\mathbf{\Delta}_s = \sum_x \Delta_x u^x$ and $\mathbf{\Delta}_s^\dagger = \sum_x \Delta_x^\dagger u^x$ we find the Fourier transform as

$$\begin{aligned} \mathcal{F}(\mathbf{\Delta}_s\phi)(k) &= \sum_{y,x} e^{-iky} \, \Delta_x u^x \phi(y) \\ &= \sum_x e^{-ikx} \, \Delta_x \mathcal{F}(\phi)(-k) \\ &= \Delta_s(k) \, \phi(-k) \\ \mathcal{F}(\mathbf{\Delta}_s^\dagger\psi)(k) &= \sum_{y,x} e^{iky} \, \Delta_x^\dagger u^x \psi(y) \\ &= \sum_x e^{ikx} \, \Delta_x^\dagger \mathcal{F}(\psi)(-k) = \overline{\sum_x e^{-ikx} \, \Delta_x} \mathcal{F}(\psi)(-k) \\ &= [\Delta_s(k)]^\dagger \, \psi(-k) \end{aligned}$$

If we order the space in such a form that $|\psi\rangle = \int \oplus_k (\psi(k), \phi(-k))$ one can write the Hamiltonian as block diagonal $\mathcal{H}_s = \oplus_k \mathbf{A_k}$ with

$$\mathbf{A_k} = \begin{pmatrix} h(k) & \Delta_s(k) \\ [\Delta_s(-k)]^\dagger & h(-k) \end{pmatrix} = \begin{pmatrix} h(k) & \Delta_s(k) \\ \overline{\Delta_s(-k)} & \overline{h(-k)} \end{pmatrix} \tag{1.5}$$

From here one can shift to the second quantized form and lift the operators back into Fock space by introducing the creation and annihilation operators with respect to the basis vectors of G_k.

$$a(|g\rangle\,|k\rangle) =: g_k^p \quad \text{for } |g\rangle\,|k\rangle \in G_k \qquad\qquad a(|g\rangle\,|k\rangle) =: g_k^h \quad \text{for } |g\rangle\,|k\rangle \in \overline{G}_k$$
$$\mathbf{a}_k := (g_k^p)_{g\in G} \qquad\qquad \mathbf{b}_k := (g_k^h)_{g\in G}$$

As the matrices $h(k)$, $\Delta(k)$ are already written in the basis of G one can directly transfer to Fock space by

$$\mathcal{H} = \int_{\Pi_d}^{\oplus} dk\, \left[(\mathbf{a}_k^\dagger)^t h(k)\, \mathbf{a}_k + (\mathbf{b}_k^\dagger)^t \overline{h}(k)\, \mathbf{b}_k + (\mathbf{a}_k^\dagger)^t \Delta(k)\, \mathbf{b}_k + (\mathbf{b}_k^\dagger)^t \Delta^\dagger(k)\, \mathbf{a}_k\right]$$

1.2.4. Winding Number

In one dimension the topological index of the bulk as in Section 1.2.2, for the case that the Hamiltonian is chiral symmetric, is the winding number also called odd chern number. The starting point of such characterisation is usually a quadratic fermionic operator [3]

$$\mathcal{H} = (\mathbf{a}^\dagger)^t\, H\, \mathbf{a}$$

with a vector of fermionic annihilation operators $\mathbf{a}$. The investigation of symmetries and the determination of topological indices can be done from the matrix H, which is assumed to be irreducible and might be called Hamiltonian matrix or shortly Hamiltonian in the following.

Fermionic systems

A chiral symmetric Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1 \quad \Gamma\Gamma^\dagger = 1. \tag{1.6}$$

The requirements on Γ ensure that it has as eigenvalues only $\{1, -1\}$ which distinguish two sublattices. In the case of equally large sublattices one can choose a basis in which Γ is

$$\Gamma = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}. \tag{1.7}$$

In this basis the Hamiltonia takes the form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \tag{1.8}$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system.

If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \tag{1.9}$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy-plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy-plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \to \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z}\, dz = \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \, \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [4, 5]

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \, \text{Tr}\left[h^{-1}(k)\partial_k h(k)\right].$$

Diagonalizing h as $h = S\Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, ...)$ and the invertible matrix S, the trace can be rewritten as

$$\begin{aligned}
\text{Tr}\left[h^{-1}\partial_k h\right] &= \text{Tr}\left[S\Lambda^{-1}S^{-1}\partial_k(S\Lambda S^{-1})\right] \\
&= \text{Tr}\left[\Lambda^{-1}\partial_k\Lambda + S^{-1}\partial_k S + (\partial_k S^{-1})S\right] \\
\text{Tr}\left[S^{-1}\partial_k S + S\partial_k S^{-1}\right] &= \text{Tr}\left[\partial_k\left(SS^{-1}\right)\right] = 0, \\
\Rightarrow \text{Tr}\left[h^{-1}\partial_k h\right] &= \text{Tr}\left[\Lambda^{-1}\partial_k\Lambda\right] = \sum_i \frac{\partial_k\lambda_i}{\lambda_i} = \partial_k\left(\sum_i \ln(\lambda_i)\right) = \partial_k \ln(\det[h]).
\end{aligned}$$

Another useful definition of the winding number is through the so-called Q-matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, ..., N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma|\psi_j\rangle = |\psi_{-j}\rangle$. The Q-matrix is defined as

$$Q = \sum_{j=1}^{N} \left(|\psi_j\rangle\langle\psi_j| - |\psi_{-j}\rangle\langle\psi_{-j}|\right).$$

As a consequence of the chiral symmetry, the Q-matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned}
\Gamma Q\Gamma &= \sum_{j=1}^{N} \left(|\psi_{-j}\rangle\langle\psi_{-j}| - |\psi_j\rangle\langle\psi_j|\right) = -Q \\
\Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}
\end{aligned}$$

The fact that the winding number of the q-matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned}
Q^2 = 1 \quad &\Rightarrow \quad qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\
QHQ = H \quad &\Rightarrow \quad qh^\dagger q = h^\dagger
\end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned}
0 &= \text{Tr}\left[q^\dagger(h^\dagger)^{-1}\partial_k(h^\dagger q)\right] \\
&= \text{Tr}\left[q\, q^\dagger\,(h^\dagger)^{-1}\partial_k h^\dagger + h^\dagger(h^\dagger)^{-1}\, q^\dagger\partial_k q\right] \\
\Rightarrow \text{Tr}\left[q^{-1}\partial_k q\right] &= -\text{Tr}\left[(h^\dagger)^{-1}\partial_k h^\dagger\right] \\
&= \text{Tr}\left[h^{-1}\partial_k h\right]
\end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of to N-component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |h_{1i}\rangle \\ |h_{2i}\rangle \end{pmatrix} \quad i \in \{1, \ldots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |h_{1i}\rangle \\ -|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle\langle h_{2i}| \\ |h_{2i}\rangle\langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow \quad q &= \sum_i |h_{1i}\rangle\langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \mathrm{Tr}[q^\dagger \partial_k q] &= \mathrm{Tr}\left[\sum_{i,j} |h_{2i}\rangle\langle h_{1i}|\,\partial_k\,(|h_{1j}\rangle\langle h_{2j}|)\right] \\ &= \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle\langle h_{1i}|\,(\partial_k |h_{1j}\rangle)\,\langle h_{2j}|\right] + \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle\langle h_{1i}|\,|h_{1j}\rangle\,(\partial_k \langle h_{2j}|)\right] \\ &= \sum_i \langle h_{1i}|\,(\partial_k |h_{1i}\rangle) + \sum_i (\partial_k \langle h_{2i}|)\,|h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\mathrm{BZ}} dk\ (\partial_k \langle h_{2i}|)\,|h_{2i}\rangle = -\int_{\mathrm{BZ}} dk\ \langle h_{2i}|\,(\partial_k |h_{2i}\rangle)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\, \mathrm{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i}\sum_i \int_{\mathrm{BZ}} dk\ (\langle h_{1i}|\,\partial_k\,|h_{1i}\rangle - \langle h_{2i}|\,\partial_k\,|h_{2i}\rangle) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$\begin{aligned} w &= \frac{1}{\pi i}\sum_i \int_{\mathrm{BZ}} dk\ \langle \Gamma\psi_i^-|\,\partial_k\,|\psi_i^-\rangle \\ &= \frac{1}{\pi i}\sum_i \int_{\mathrm{BZ}} dk\ \langle \psi_i^+|\,\partial_k\,|\psi_i^-\rangle \\ &= \frac{1}{2\pi i}\sum_i \int_{\mathrm{BZ}} dk\ (\langle h_{1i}|\,\partial_k\,|h_{1i}\rangle - \langle h_{2i}|\,\partial_k\,|h_{2i}\rangle) \end{aligned}$$

The quantity $\sum_i \langle \Gamma\psi_i^-|\,\partial_k\,|\psi_i^-\rangle$ can be related to a polarisation [6].

Topological indices for bosonic systems

For bosonic systems a topological analysis can also be performed for quadratic BdG Hamiltonians [2]. In this case one also resorts to the single-particle Hamiltonian, in which there is in principle no distinction between fermions and bosons. However the dynamics of the intire system are of course bosonic, which has to be somehow incorporated into the single-body analysis. For this purpose one considers the matrix form of the BdG Hamiltonian.

$$\mathcal{H} = \frac{1}{2}\begin{pmatrix} (\mathbf{a}^\dagger)^t & \mathbf{a}^t \end{pmatrix} \tilde{A} \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix}, \qquad \tilde{A} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & \overline{h} \end{pmatrix}$$

The equations of motion for the annihilation operators are given by

$$\frac{d}{dt}\,\mathbf{a} = i\,[\mathcal{H}, \mathbf{a}] = -i\,H\,\mathbf{a}, \tag{1.10}$$

with dynamical matrix

$$H = \begin{pmatrix} h & \Delta \\ -\Delta^\dagger & -\overline{h} \end{pmatrix}$$

As argued in [2] it is the eigenenergies and eigenstates of this matrix that determine the propagation of the system and hence the topological indices have to be computed with respect to this matrix, or more precisely for the single-particle Hamiltonian

$$\mathcal{H}_s = \begin{pmatrix} \mathbf{h} & \boldsymbol{\Delta} \\ -\boldsymbol{\Delta}^\dagger & -\overline{\mathbf{h}} \end{pmatrix}$$

In order to compute the winding number one proceeds like in the fermionic case and use the formalism for the matrix H, which can be generated by the operation

$$H = JA \qquad \text{with } J = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} \tag{1.11}$$

H is not self-adjoint and has not necessarily real eigenvalues. Due to Eq. (1.10) imaginary eigenvalues can lead to non-stable dynamics. Therefore one is restricted to dynamical matrices H with real spectrum in order to perform a topological analysis. Eq. (1.11) can also be stated as H being J-self-adjoint on a Krein space [2], i.e.

$$H^\dagger J = AJ^2 = A = JH$$

One can now define a winding number for the case of H being chiral symmetric, i.e. there is unitary transformation U into a certain basis where H is block-off diagonal.

$$UH_kU^\dagger = \begin{pmatrix} 0 & S \\ T & 0 \end{pmatrix} =: H'$$

The J matrix transforms under U to J', which is still hermitian and squares to the identity. Setting

$$J' = \begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix}$$

one can write the property of J-self-adjointness explicitly as

$$\begin{aligned} H'^\dagger J' &= J' H' \\ \Leftrightarrow \quad \begin{pmatrix} T^\dagger B^\dagger & T^\dagger C \\ S^\dagger A & S^\dagger B \end{pmatrix} &= \begin{pmatrix} BT & AS \\ CT & B^\dagger S \end{pmatrix} \\ \Leftrightarrow \quad S &= A^{-1}T^\dagger C \\ T &= B^{-1}T^\dagger B^\dagger \\ S &= (B^\dagger)^{-1}S^\dagger B \end{aligned}$$

The equation $S = A^{-1}T^\dagger C$ gives the insight that calculating the winding with S or T is equivalent. This becomes clear when one remembers the definition of the winding number for a hermitian block-off-diagonal matrix with blocks h and $h^\dagger$, i.e.

$$w = \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\, \mathrm{Tr}\left[h^{-1}(k)\partial_k h(k)\right].$$

Investigating the kernel of the integral when replacing h by S, one can put the winding of the two blocks in relation.

$$\begin{aligned} \mathrm{Tr}\left[S^{-1}\partial_k S\right] &= \mathrm{Tr}\left[C^{-1}(T^\dagger)^{-1}A\,\partial_k\left(A^{-1}T^\dagger C\right)\right] \\ &= \mathrm{Tr}\left[(T^\dagger)^{-1}\partial_k T^\dagger\right], \end{aligned}$$

as A and C are independent of k. This results in a winding number of $w(S) = -w(T)$, which is the same relation as between the hermitian blocks ($w(h) = -w(h^\dagger)$). Consequently the winding number is well defined.

2. Topological Classification of motional modes in Rydberg atom arrays

The possibility of trapping atoms in optical tweezers and arranging them in arbitrary geometries has inspired the investigation of various topological properties in such arrays. One of the targets of such investigations has been the motional modes of the atoms in the array. As the atoms move in the harmonic oscillator potential of the tweezer, the interaction between atoms, which are distance dependent, also change. This leads to a coupling between the motional modes of the atoms. In this work the atoms will occupy Rydberg states leading to dipole-dipole or van der Waals interactions. There are two regimes of interest, which behave in a different way. If the trap depth is much larger than the interaction, the generation of additional motional excitations is suppressed, and only hopping of motional excitations between sites takes place. Thus topological classification can exploit the single particle property of the dynamics. In the complementary regime the excitation number is not conserved and one has to deploy a bosonic formalism for the topological classification, via the Bogoliubov de Gennes (BdG) picture, discussed in the previous chapter. In the next section it is briefly discussed why both regimes can be implemented in Rydberg atom arrays.

2.1. Parameter regimes

Important for the presence or absence of motional excitation number conservation is the ratio of interaction strength V to trap depth ω. The lifetime of the Rydberg states T, however, sets a lower bound on the interaction strength, as the dynamics induced by the interaction has to be faster than the lifetime of the Rydberg state, in order to be observed. For the regime, in which the motional excitation number is conserved, the trap depth has to be much larger than the interaction which in turn has to be much larger than this lower bound. In the following I will discuss the relevant energy scales and show that both regimes can be implemented in Rydberg atom arrays.

The lifetimes for conventional Rydberg states can reach up to several hundred μs. For Circular Rydberg states the lifetimes can reach even a few ms. In both cases the trap depth (ω) can easily have values of a few MHz [7, 8]. Correspondingly the lower bound for the interaction strength is located at $V >> 1/T \sim 1\,\text{kHz}$. With the trap depth being in the MHz range, the conditions of conserved motional excitation number can be easily met, as the interactions between Rydberg atoms can be tuned in wide range from kHz to GHz. Due to this large flexibility in the interaction strength, also the complementary regime of non-conserved motional excitation number can be realized. In the following I will derive the effective Hamiltonian for the motional modes.

2.2. Model

For the topological analysis a model will be used, that only takes motional degrees of freedom into account. This can be achieved by assuming that all atoms are in the same internal state, and the interaction doesn't couple different states. The interaction that results by this is a van der Waals interaction for atoms in Rydberg states or a dipole-dipole interaction if the atoms are dressed with an external field. I will keep the form of the interaction general in the following derivation. The system consists of N atoms, which are trapped in optical tweezers with centers $\{\mathbf{R}_l\}_{l\in\{1,\ldots,N\}}$. The Hamiltonian that describes the model consists of the harmonic trap potential part $\mathcal{H}_0$ and an interaction part $\mathcal{H}_I$, that depends on the position of the atoms.

$$
\begin{aligned}
\mathcal{H} &= \mathcal{H}_0 + \mathcal{H}_I(\{\mathbf{R}_l\}) \\
\text{with} \quad \mathcal{H}_I(\{\mathbf{R}_l\}) &= \frac{1}{2}\sum_{i,j} V_{i,j}(\{\mathbf{R}_l\})\, n_i n_j \\
\text{where} \quad V_{i,j}(\{\mathbf{R}_l\}) &= V(\mathbf{R}_i, \mathbf{R}_j)
\end{aligned}
$$

The operators n_i are the projectors on the internal state of the ith atom. As all atoms are assumed to be in the same internal state, the n_i become redundant and can be dropped. The interaction Hamiltonian then reads

$$
\mathcal{H}_I(\{\mathbf{R}_l\}) = \sum_{i,j;\, i>j} V_{i,j}(\{\mathbf{R}_l\})
$$

The $V_{i,j}$ are formally functions of all positions $\{\mathbf{R}_l\}$, i.e $V_{i,j}(\{\mathbf{R}_l\})$. But of course they only depend on $\mathbf{R}_i$ and $\mathbf{R}_j$, i.e.

$$
\frac{\partial V_{i,j}}{\partial \mathbf{R}_l} = 0 \quad \text{if}\, i \neq l \neq j
$$

The displacements of the atoms from the tweezer centers $\{\delta\mathbf{R}_l\}$ are assumed to be small compared to the distance between the tweezers, which justifies a Taylor expansion of the interaction potential to second order. As shown in Appendix A this leads to an interaction Hamiltonian of the form

$$
\mathcal{H}_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i}\, \delta\mathbf{R}_i + \sum_{i,j} U_{i,j}\, \delta\mathbf{R}_i\, \delta\mathbf{R}_j
$$

with

$$
\begin{aligned}
U_{i,j} &= \frac{1}{2}\, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \quad \text{if}\, i \neq j \\
U_{i,i} &= \frac{1}{2} \sum_{j;\, j\neq i} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2}
\end{aligned}
\tag{2.1}
$$

In order to allow for a quantum mechanical description of the displacements $\delta\mathbf{R}_l$ are treated as operators.

2.3. No particle number conservation

Even for a bosonic system a topological classification with respect to the Altland-Zirnbauer symmetry classes is possible. As discussed in Section 1.2.3, the concept of holes has to be introduced by rewriting the Hamiltonian in a BdG form. For this it is, however, necessary that the Hamiltonian is quadratic in the displacements. The linear term originating from the first order expansion of the interaction potential however breaks this condition. In order to remove this term, the trap centers can be redefined, such that the linear term vanishes. For this purpose the displacement operators are shifted by constant vectors $\{\delta\mathbf{R}_l\} = \{\delta\tilde{\mathbf{R}}_l + \mathbf{\Delta}_l\}$. In the following I will treat the 1D case and the interaction is assumed to be of the dipole-dipole type, i.e.

$$
\begin{aligned}
V_{dd}(\mathbf{r}) &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\left(1 - 3\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right) =: V_0(r)\, f(\hat{\mathbf{r}}) = V_0(a)\,\frac{a^3}{r^3}\, f(\hat{\mathbf{r}}) \\
\frac{\partial V_{dd}}{\partial r_\alpha} &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\,\frac{1}{r}\left(-3\,\frac{r_\alpha}{r} + 15\,\frac{r_\alpha}{r}(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2 - 6\,\hat{d}_\alpha\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})\right) =: \frac{V_0(r)}{r}\, g(\hat{\mathbf{r}}) = \frac{V_0(a)}{a}\,\frac{a^4}{r^4}\, g(\hat{\mathbf{r}}) \\
\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\,\frac{1}{r^2}\left[\delta_{\alpha\beta}\left(-3 + 15\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right) - 6\,\hat{d}_\alpha\hat{d}_\beta + 30\,\frac{\hat{d}_\alpha r_\beta + \hat{d}_\beta r_\alpha}{r}\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}}) + 15\,\frac{r_\alpha r_\beta}{r^2}\left(1 - 7\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right)\right] \\
&=: \frac{V_0(r)}{r^2}\, c(\hat{\mathbf{r}}) = \frac{V_0(a)}{a^2}\,\frac{a^5}{r^5}\, c(\hat{\mathbf{r}})
\end{aligned}
$$

In the above equations d is the dipole operator, a is the distance between the tweezers and $\mathbf{r} = \mathbf{R}_2 - \mathbf{R}_1$ is the distance vector between two atoms. The hat symbol denotes vectors with unit norm. In combination with the harmonic trap potential this leads to a Hamiltonian for the shifted displacements $\{\delta\tilde{\mathbf{R}}_l\}$ of the form

$$
\begin{aligned}
\mathcal{H}(\{\delta\tilde{\mathbf{R}}_l\}) &= \sum_i \frac{\mathbf{P}_i}{2m} + \sum_i \frac{1}{2} m\omega^2 \left(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i\right)^2 + \sum_i \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \left(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i\right) + \sum_{i,j} \left(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i\right) U_{i,j} \left(\delta\tilde{\mathbf{R}}_j + \boldsymbol{\Delta}_j\right) \\
&= \sum_i \frac{\mathbf{P}_i}{2m} + \sum_i \frac{1}{2} m\omega^2\, \delta\tilde{\mathbf{R}}_i^2 + \sum_{i,j} \delta\tilde{\mathbf{R}}_i\, U_{i,j}\, \delta\tilde{\mathbf{R}}_j + \sum_i \left(\frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j \left(2\, U_{j,i} + m\omega^2\, \delta_{ij}\right) \boldsymbol{\Delta}_j \right) \delta\tilde{\mathbf{R}}_i,
\end{aligned}
$$

where $\mathbf{P}_i$ are the momentum operators of the atoms. In the second line all constant terms have been dropped.

Thus the linear term can be removed by choosing the shift vectors $\{\boldsymbol{\Delta}_l\}$ such that

$$
\left(\frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j \left(2\, U_{j,i} + m\omega^2\, \delta_{ij}\right) \boldsymbol{\Delta}_j \right) = 0 \quad \forall i
$$

By defining the following dimensionless quantities

$$
\begin{aligned}
\tilde{g}_i &:= \frac{a}{V_0(a)} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \\
U'_{i,j} &:= \frac{a^2}{V_0(a)}\, 2\, U_{j,i}
\end{aligned}
$$

the condition for the shift can be written as

$$
\frac{V_0(a)}{a} G_i := \frac{V_0(a)}{a} \left(\tilde{g}_i + \sum_j W_{ij} \frac{\Delta_j}{a} \right) := \frac{V_0(a)}{a} \left(\tilde{g}_i + \sum_j \left(U'_{j,i} + \frac{m\omega^2 a^2}{V_0(a)} \delta_{ij} \right) \frac{\Delta_j}{a} \right) = 0 \quad \forall i
$$

Here $\tilde{g}$ and U' are dimensionless and the the term $m\omega^2 a^2/[V_0(a)\max(U')] =: m\omega^2/V_{\max}$ chosen of order 0.01-10. Now I use the two free parameters h and the angle φ_d of the dipole moment $\hat{\mathbf{d}}$ to set G_i close to zero for different values of b. Close to zero means here that $\max(|G_i|)$ compared to $\sqrt{\hbar/(2m\omega)}\max(U')/a$ is small. Here $\sqrt{\hbar/(2m\omega)}/a \sim 10^{-3}$ is the displacement from a single motional excitation in units of a.

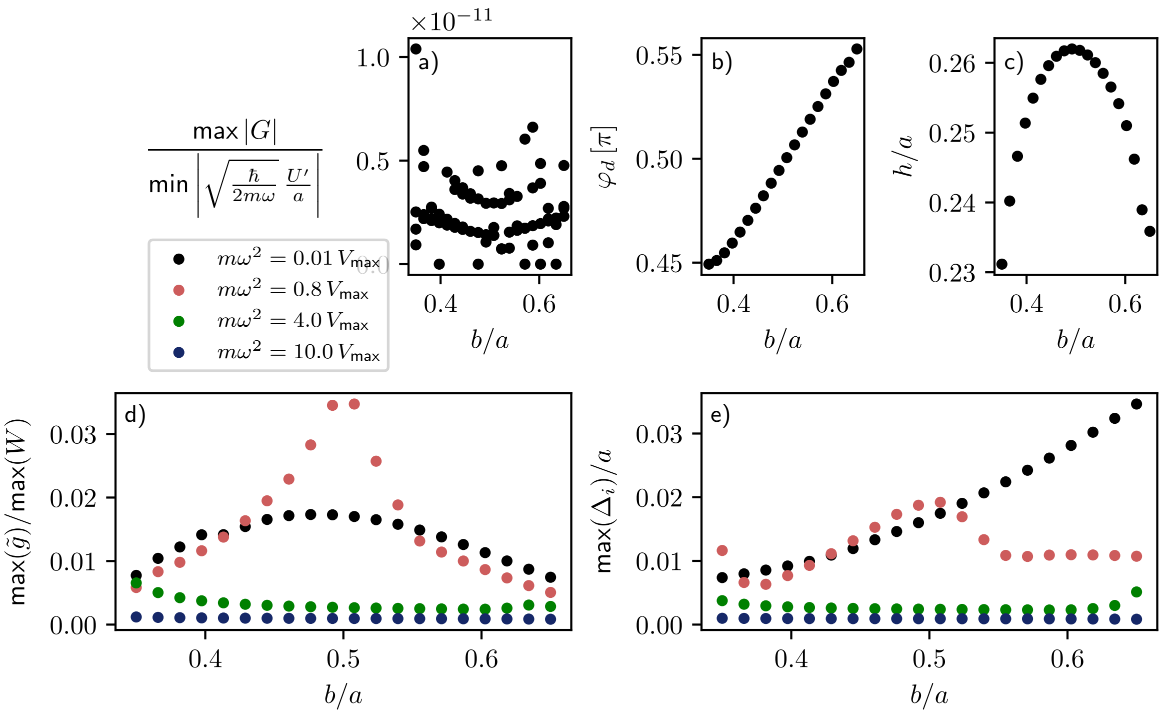

Figure 2.1: **Parameters for negligible first order.** For different values of $m\omega^2$ the parameters h and φ_d are tuned such that the first order is negligible. The color code indicates the value of $m\omega^2/V_{\max}$. The resulting ratio between first and second order is shown in panel a) for all considered values of $m\omega^2/V_{\max}$ 0.01, 0.8, 4 and 10. As an example the optimal parameters h and φ_d for $m\omega^2/V_{\max} = 4$ are shown in panel b) and c). The maximal shift of the trap center necessary to eliminate the first order is shown in panel e), while panel d) shows that size of the shift is mainly determined by the ratio between $\tilde{g}$ and W. For the calculations a chain length of 8 atoms was used.

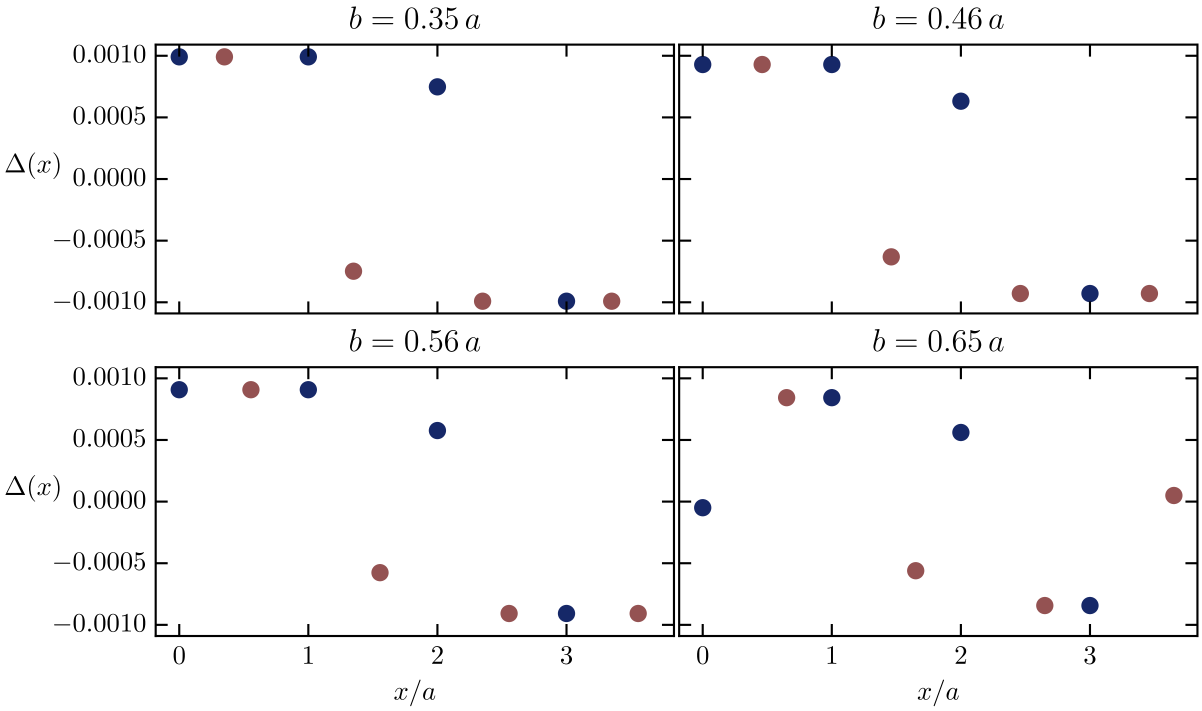

Figure 2.2: **Trap center shift.** For $m\omega^2 = 10\,V_{\max}$ the shift of the trap center for each site, that tunes the linear order to zero, is shown. The length of the chain is taken as 8 atoms.

Fig. 2.1 and Fig. 2.2 tell us that the Δ-displacements are of order 10^{-2}, an order of magnitude bigger than the single motional mode displacement. Nevertheless the shift is small compared to the atomic distance a, which allows us to stick to the assumption of a periodic lattice. The shifts get smaller with increasing trap frequency. By eliminating the linear order, the Hamiltonian takes a quadratic form in the displacement and can hence be written as

$$\mathcal{H} = \sum_i \frac{\mathbf{P}_i}{2m} + \sum_i \frac{1}{2} m\omega^2\, \delta\tilde{\mathbf{R}}_i^2 + \sum_{i,j} \delta\tilde{\mathbf{R}}_i\, U_{i,j}\, \delta\tilde{\mathbf{R}}_j$$

In second quantized from where I write the momentum and displacement operators in terms of creation and annihilation operators

$$\delta\tilde{\mathbf{R}}_i = \sqrt{\frac{\hbar}{2m\omega}}\,(a_i + a_i^\dagger)$$
$$\mathbf{P}_i = i\sqrt{\frac{\hbar m\omega}{2}}\,(a_i^\dagger - a_i)$$

the total Hamiltonian reads

$$\begin{aligned}\mathcal{H} &= \sum_i \hbar\omega\, a_i^\dagger a_i + \frac{\hbar}{2m\omega} \sum_{i,j} U_{i,j}\, \left(a_i\, a_j + a_i^\dagger a_j + a_i\, a_j^\dagger + a_i^\dagger a_j^\dagger\right) \\ &= \mathbf{a}^\dagger\, A\, \mathbf{a}\end{aligned}$$

with $\mathbf{a} = (a_1, \dots, a_N, a_1^\dagger, \dots, a_N^\dagger)$ the vector of operators and a coupling matrix A, which establishes the BdG form of the Hamiltonian and has itself the following structure

$$A = \frac{\hbar}{2m\omega}\begin{pmatrix}\bar{U} & U \\ U & \bar{U}\end{pmatrix}$$
$$\text{with} \quad \bar{U} = U + m\omega^2.$$

The dynamical matrix is then given by

$$H = \frac{\hbar}{m\omega} \begin{pmatrix} \bar{U} & U \\ -U & -\bar{U} \end{pmatrix}$$

As it is a real matrix, it is time reversal symmetric. Particle-hole symmetry is also fulfilled, as $\overline{H} = -\Sigma_x H \Sigma_x$ with

$$\Sigma_x = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}$$

Consequently it is then also chiral symmetric, as $H = -\Sigma_x H \Sigma_x$. This puts the system in the BDI symmetry class, which allows for a topological characterization in one dimension. Since H is not hermitian one has to investigate its spectrum for dynamical stability, i.e. no complex eigenenergies are allowed. Fig. 2.3 shows that for small trapping depth compared to the phonon interactions always large imaginary contributions to the eigenvalues are observed. But for large tweezer strength ($m\omega^2 = 10\, V_{\max}$), the imaginary part of the spectrum vanishes and also eigenstates inside of a gap are visible.

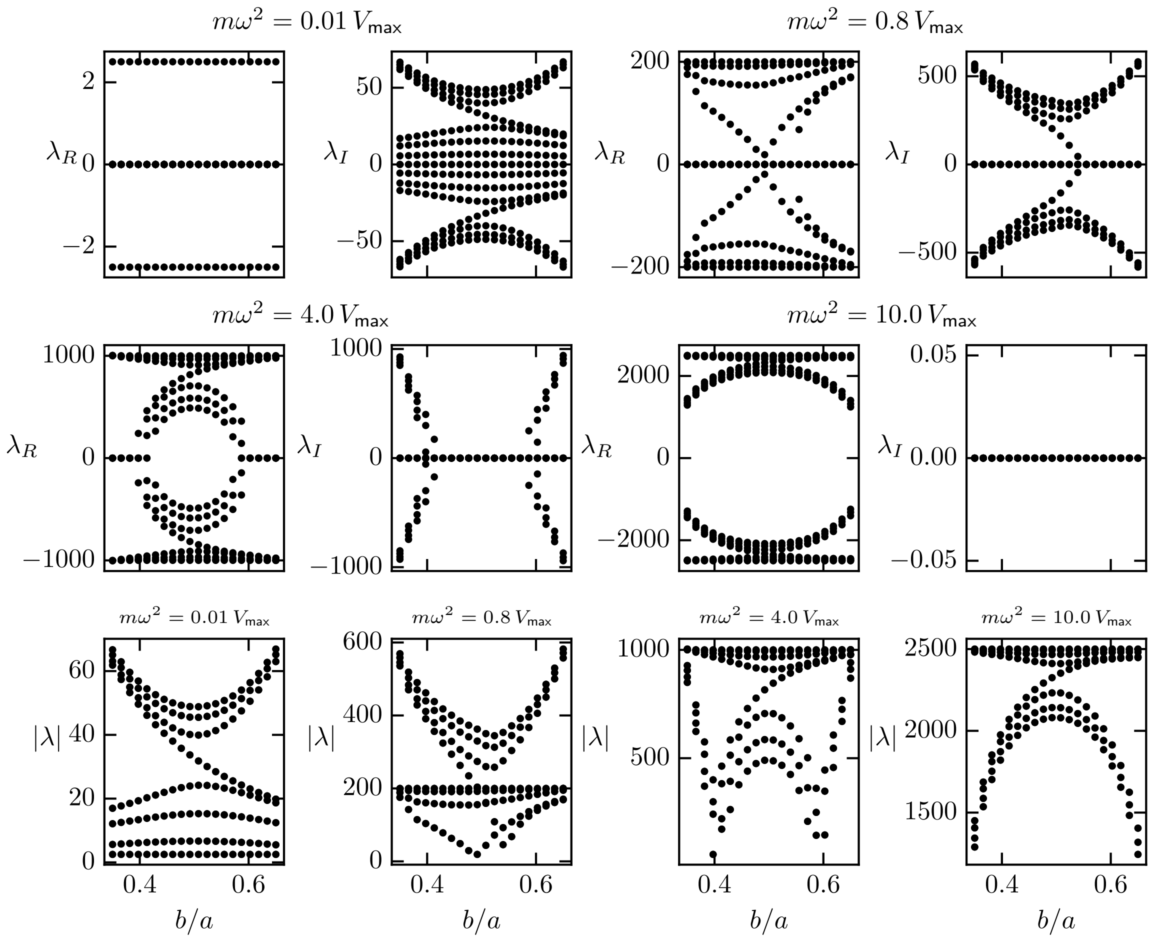

Figure 2.3: **Spectrum of BdG Hamiltonian for different trap frequencies.** For different strengths of the optical trap the real and imaginary part of the spectrum is plotted. The atom number used for this computation is 8. In the last row the absolute value of the spectra is shown.

However by calculating the positive spectrum for larger atom numbers, as depicted in Fig. 2.4 one realizes that there is no real gap that withstands an infinite lattice size limit. This indicates that the system does not possess topological states, which can be affirmed by calculating the system's winding number, under the approximation that the periodicity breaking due to the small shifts of the trap center can be neglected.

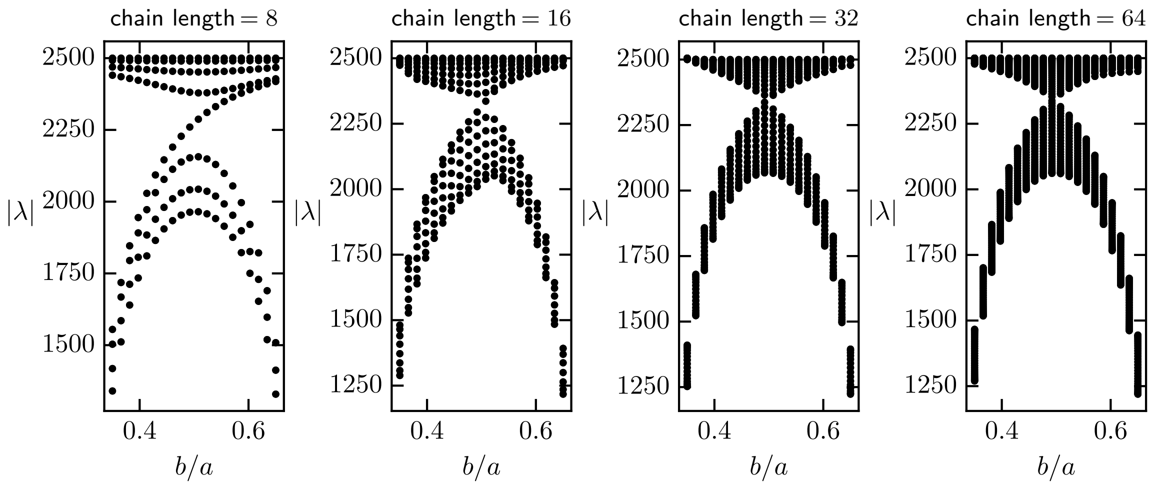

Figure 2.4: **Positive spectrum of BdG Hamiltonian for different atom numbers.**

The first step is to transform the dynamical matrix into momentum space. When recapitulating the momentum space form of the general Bogoliubov-de Gennes Hamiltonian, Eq. (1.5), one can infer the structure of the dynamical matrix as

$$H_k = \begin{pmatrix} \tilde{U}(k) & U(k) \\ -\overline{U(-k)} & -\overline{\tilde{U}(-k)} \end{pmatrix} = \begin{pmatrix} \tilde{U}(k) & U(k) \\ -U(k) & -\tilde{U}(k) \end{pmatrix}$$

This form is independent of the explicit shape of U and the potential. By assuming only that U is generated by a second order expansion of a potential with two sublattices, i.e. Eq. (2.1), the form of H_k can be inferred.

$$H_k = \begin{pmatrix} \tilde{U}(k) & U(k) \\ -U(k) & -\tilde{U}(k) \end{pmatrix}$$
$$\tilde{U}_k = \begin{pmatrix} b(k) & d(k) \\ \bar{d}(k) & b(k) \end{pmatrix} \quad U_k = \begin{pmatrix} c(k) & d(k) \\ \bar{d}(k) & c(k) \end{pmatrix}$$

Exploiting the symmetric structure of the matrix one can write

$$\begin{aligned} & H_k = \tau_z \otimes \tilde{U}_k + i\tau_y \otimes U_k \\ & \tilde{U}_k = \sigma_0\, b(k) + \sigma_x \operatorname{Re}[d(k)] - \sigma_y \operatorname{Im}[d(k)] \\ & U_k = \sigma_0\, c(k) + \sigma_x \operatorname{Re}[d(k)] - \sigma_y \operatorname{Im}[d(k)] \\ \Rightarrow \quad & H_k = b(k)\, \tau_z \otimes \sigma_0 + c(k)\, i\tau_y \otimes \sigma_0 + \operatorname{Re}[d(k)]\,(\tau_z + i\tau_y) \otimes \sigma_x - \operatorname{Im}[d(k)]\,(\tau_z + i\tau_y) \otimes \sigma_y \end{aligned}$$

where τ_α, σ_α $\alpha \in \{0, x, y, z\}$ are the usual Pauli matrices. In these expressions b, c, d can be matrices treating the different directions of displacements from the trap center. The tensor product is not written explicitly in these cases. Writing an arbitrary matrix in the form

$$X = \mathbb{1} \otimes A + \tau_x \otimes B + \tau_y \otimes C + \tau_z \otimes D$$

one can compute the commutation relation by using

$$[O \otimes P, R \otimes S] = \frac{1}{2}\,[O, R] \otimes \{P, S\} + \frac{1}{2}\,\{O, P\} \otimes [R, S]$$

$$
\begin{aligned}
[H_k, X] &= \left[\left(\tau_z \otimes \tilde{U} + i\tau_y \otimes U\right)\ ,\ \left(\mathbb{1} \otimes A + \tau_x \otimes B + \tau_y \otimes C + \tau_z \otimes D\right)\right] \\
&= \tau_z \otimes [\tilde{U}, A] + i\tau_y \otimes [U, A] + i\tau_y \otimes \{\tilde{U}, B\} + \tau_z \otimes \{U, B\} \\
&\quad - i\tau_x \otimes \{\tilde{U}, C\} + i\mathbb{1} \otimes [U, C] + \mathbb{1} \otimes [\tilde{U}, D] - \tau_x \otimes \{U, D\} \\
&= \mathbb{1} \otimes (i[U, C] + [\tilde{U}, D]) - \tau_x \otimes (i\{\tilde{U}, C\} + \{U, D\}) \\
&\quad + \tau_y \otimes (i[U, A] + i\{\tilde{U}, B\}) + \tau_z \otimes ([\tilde{U}, A] + \{U, B\})
\end{aligned}
$$

All of the summands are independent of each other, as are the pauli matrices. Therefore every summand has to vanish itself. To show that this is true I use the similarity of $\tilde{U}$ and U, in the form $\tilde{U} = \mathbb{1} \otimes b(k) + W$ and $U = \mathbb{1} \otimes c(k) + W$. The vanishing of the terms that come with $\mathbb{1}$ and τ_x yields

$$
\begin{aligned}
i\,[U, C] + [\tilde{U}, D] &= 0 \\
\Rightarrow \quad i\,[\mathbb{1} \otimes c(k), C] + [\mathbb{1} \otimes b(k), D] + [W, i\,C + D] &= 0 \\
\text{and} \quad i\,\{\tilde{U}, C\} + \{U, D\} &= 0 \\
\Rightarrow \quad i\,\{\mathbb{1} \otimes c(k), C\} + \{\mathbb{1} \otimes b(k), D\} + \{W, i\,C + D\} &= 0
\end{aligned}
$$

$b(k)$ and $c(k)$ differ only by a constant diagonal term, originating from the trap potential. This lets the last row read

$$
i\,\frac{\omega}{2}\,\{\mathbb{1}, C\} + \{\mathbb{1} \otimes b(k), i\,C + D\} + \{W, i\,C + D\} = 0
$$

By exploiting that this equation has to hold for any k with X k-independent and realizing that $b(k)$ and $W(k)$ are independent functions one follows that $C = 0$ and therefore also $D = 0$. Similarly one proceeds for the first and second term

$$
\begin{aligned}
[U, A] + \{\tilde{U}, B\} &= 0 \\
\Rightarrow \quad [\mathbb{1} \otimes c(k), A] + \{\mathbb{1} \otimes b(k), B\} + [W, A] + \{W, B\} &= 0 \\
[\tilde{U}, A] + \{U, B\} &= 0 \\
\Rightarrow \quad [\mathbb{1} \otimes b(k), A] + \{\mathbb{1} \otimes c(k), B\} + [W, A] + \{W, B\} &= 0 \\
\Rightarrow \quad -\frac{\omega}{2}\,[\mathbb{1}, A] + \frac{\omega}{2}\,\{\mathbb{1}, B\} &= 0 \\
\Rightarrow \quad B &= 0
\end{aligned}
$$

Using again the independence of $b(k)$ and $W(k)$ one follows $[W, A] = 0$. The matrix W can then be written as

$$
\begin{aligned}
W &= \sum_{\alpha,\beta \in \{0,x,y,z\}} w_{\alpha,\beta}(k)\,\sigma_\alpha \otimes \sigma_\beta \\
&= \sigma_x \otimes (w_{x,0}\,\mathbb{1} + w_{x,x}\,\sigma_x + w_{x,z}\,\sigma_z) + \sigma_y \otimes (w_{y,0}\,\mathbb{1} + w_{y,x}\,\sigma_x + w_{y,z}\,\sigma_z)
\end{aligned}
$$

Due to the independence of the $w_{\alpha,\beta}(k)$ functions, A has to commute with every summand individually. The only matrix which is not proportional to the identity, that commutes with the σ_x-term is $\sigma_x \otimes \mathbb{1}$. But this does not commute with the σ_y-term. Hence the A has to be proportional to the identity and therefore X has to be proportional to the identity.

Thus H_k is irreducible as it has no unitary symmetries. This allows to perform a topological classification. Exploiting the chiral symmetry, one can change to a basis where the chiral transformation $\Sigma_x^k = (\mathbb{1}, -\mathbb{1})$. In this basis H_k takes the form

$$
H_k' = \begin{pmatrix} 0 & S_k \\ T_k & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \mathbb{1} & \mathbb{1} \\ \mathbb{1} & -\mathbb{1} \end{pmatrix} H_k \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ \mathbb{1} & -\mathbb{1} \end{pmatrix}
$$

$$
S_k = \begin{pmatrix} b(k) - c(k) & 0 \\ 0 & b(k) - c(k) \end{pmatrix} \quad T_k = \begin{pmatrix} b(k) + c(k) & 2\,d(k) \\ 2\,\bar{d}(k) & b(k) + c(k) \end{pmatrix}
$$

As both matrices S_k and T_k are hermitian the Hamiltonian has vanishing winding number, indicating a topological trivial phase. The structure of S_k and T_k remains the same for higher dimensional displacements. I want to mention that if there are no pairing terms, i.e. $\mathbf{\Delta} = 0$, the Hamiltonian is block diagonal and therefore not irreducible. In this case the introduction of holes is redundant and the winding number can be calculated from a single block, which can lead to non-vanishing winding numbers. It is thereby shown that the pairing terms prohibit the occurrence of topologically non-trivial phases in 1D for motional degrees of freedom stemming from a second order tailor expansion of an interaction potential.

2.4. Excitation conservation

In this section I want to discuss the case, where the trapping strength dominates over the the interaction between the atoms, i.e. $m\omega^2 >> \max(U)$. In this regime the creation of additional excitations through the interaction channel is suppressed and can be neglected. As a consequence the Hamiltonian can be projected onto a subspace of fixed excitation number, which is equivalent to a first order Schrieffer-Wolff transform. The result is that the terms that change the motional excitation number vanish and the Hamiltonian takes the form

$$\begin{aligned}\mathcal{H} &= \sum_i \hbar\omega\, a_i^\dagger a_i + \frac{\hbar}{2m\omega} \sum_{i,j} U_{i,j} \left(a_i^\dagger a_j + a_i\, a_j^\dagger \right) \\ &= \frac{\hbar}{m\omega} \sum_{i,j} \left(\delta_{i,j}\, m\omega^2 + U_{i,j} \right) a_i^\dagger a_j \\ &= (\mathbf{a}^\dagger)^t A\, \mathbf{a} \qquad \text{with } A = \frac{\hbar}{m\omega} \left(\delta_{i,j}\, m\omega^2 + U_{i,j} \right)_{i,j}\end{aligned}$$

Again the vector of annihilation operators is introduced $\mathbf{a} = (a_1, \ldots, a_N)$, this time for the particle space only.

A. Derivation of the Hamiltonian for the chain of Rydberg atoms in optical tweezers

As described in the main text, the assumption of small displacements $\{\delta \mathbf{R}_l\}$ compared to the distance between the tweezers motivates the Taylor expansion of the interaction potential to second order.

$$\begin{aligned}
\mathcal{H}_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n,m} \delta\mathbf{R}_n\,\frac{\partial^2 \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m} \delta\mathbf{R}_n\,\frac{\partial^2 \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m + \frac{1}{2}\sum_{n=m} \delta\mathbf{R}_n\,\frac{\partial^2 \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\,i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&+ \frac{1}{2}\sum_n \sum_{i,j;\,i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}\,\delta\mathbf{R}_n \\
&= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\,i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m\,(\delta_{n=i,m=j} + \delta_{n=j,m=i}) \\
&+ \frac{1}{2}\sum_n \sum_{i,j;\,i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}\,\delta\mathbf{R}_n\,(\delta_{n=i} + \delta_{n=j})
\end{aligned}$$

In order to allow for a quantum mechanical description the displacements $\delta\mathbf{R}_l$ are treated as operators. The second order term can be simplified by exploiting the symmetry of the second derivative and the fact that $V_{i,j}$ only depends on $\mathbf{R}_i$ and $\mathbf{R}_j$. Thus

$$\sum_{i,j;\,i>j}\left(\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2}\,\delta\mathbf{R}_i^2 + \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j^2}\,\delta\mathbf{R}_j^2\right) = \sum_{i,j;\,i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2}\,\delta\mathbf{R}_i^2$$

and

$$\sum_{i,j;\,i>j}\left(\delta\mathbf{R}_i\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j}\,\delta\mathbf{R}_j + \delta\mathbf{R}_j\,\frac{\partial^2 V_{j,i}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j \partial \mathbf{R}_i}\,\delta\mathbf{R}_i\right) = \sum_{i,j;\,i\neq j} \delta\mathbf{R}_i\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j}\,\delta\mathbf{R}_j$$

The interaction can therefore be written in the following form

$$\begin{aligned}
\mathcal{H}_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i}\,\delta\mathbf{R}_i + \frac{1}{2}\sum_{i,j;\,i>j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j}\,(\delta\mathbf{R}_i\,\delta\mathbf{R}_j + \delta\mathbf{R}_j\,\delta\mathbf{R}_i) \\
&+ \frac{1}{2}\sum_{i,j;\,i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2}\,\delta\mathbf{R}_i^2 \\
&= \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i}\,\delta\mathbf{R}_i + \frac{1}{2}\sum_{i,j;\,i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j}\,\delta\mathbf{R}_i\,\delta\mathbf{R}_j \\
&+ \frac{1}{2}\sum_{i,j;\,i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2}\,\delta\mathbf{R}_i^2
\end{aligned}$$

By introducing the matrix U the expanded interaction Hamiltonian takes the compact form

$$\mathcal{H}_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = \mathcal{H}_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial \mathcal{H}_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \sum_{n,m} U_{n,m}\,\delta\mathbf{R}_n\,\delta\mathbf{R}_m$$

Where the constant energy shift $\mathcal{H}_I(\{\mathbf{R}_l\})$ can be ignored and the matrix U is defined as

$$U_{n,m} = \frac{1}{2}\frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if}\, n \neq m$$
$$U_{n,n} = \frac{1}{2}\sum_{j;\,j\neq n}\frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}$$

For an interaction that only depends on the relative distance between the atoms $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$, i.e. $V_{i,j}(\{\mathbf{R}_l\}) = V_{i,j}(\mathbf{R}_i - \mathbf{R}_j)$, the matrix U can be further specified. Using the fact that

$$\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_i} = \mathbb{1} = -\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_j}$$

the derivatives in U can be written as

$$\begin{aligned}\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} &= \frac{\partial}{\partial \mathbf{R}_n}\left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}}\frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m}\right)\\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}\frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m}\frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}}\frac{\partial^2 \mathbf{r}_{nm}}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\\ &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}\end{aligned}$$

and

$$\begin{aligned}\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_m^2 m} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}\frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m}\frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m}\\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}\end{aligned}$$

Thus

$$U_{n,m} = -\frac{1}{2}\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if}\, n \neq m$$
$$U_{n,n} = \frac{1}{2}\sum_{j;\,j\neq n}\frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$\mathcal{H}_0(\{\delta\mathbf{R}_l\}) = \sum_i \frac{\mathbf{P}_i^2}{2m} + \frac{1}{2}m\omega^2\,\delta\mathbf{R}_i^2,$$

where $\mathbf{P}_i$ is the momentum of the ith atom. I can now shift the considered displacements from the trap centers by a constant $\{\delta\mathbf{R}_l\} = \{\delta\tilde{\mathbf{R}}_l + \boldsymbol{\Delta}_l\}$, where $\delta\mathbf{R}_l$ and $\delta\tilde{\mathbf{R}}_l$ are operators and $\boldsymbol{\Delta}_l$ are just constant vectors. Neglecting all constant terms the Hamiltonian minus the momentum term $\tilde{\mathcal{H}} = \mathcal{H} - \sum_i \mathbf{P}_i/2m$ up to second order reads

$$\begin{aligned}\tilde{\mathcal{H}}(\{\delta\mathbf{R}_l\}) &= \sum_i \frac{1}{2}m\omega^2\,(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i)^2 + \sum_i \frac{\partial\mathcal{H}_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i}\,(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i) + \sum_{i,j}(\delta\tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i)\,U_{i,j}\,(\delta\tilde{\mathbf{R}}_j + \boldsymbol{\Delta}_j)\\ &= \sum_i \frac{1}{2}m\omega^2\,\delta\tilde{\mathbf{R}}_i^2 + \sum_{i,j}\delta\tilde{\mathbf{R}}_i\,U_{i,j}\,\delta\tilde{\mathbf{R}}_j + \sum_i\left(\frac{\partial\mathcal{H}_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i} + \sum_j\left(2\,U_{j,i} + m\omega^2\,\delta_{ij}\right)\boldsymbol{\Delta}_j\right)\delta\tilde{\mathbf{R}}_i,\end{aligned}$$

where it was used that U is symmetric. As the matrix $\left(2\,U_{j,i} + m\omega^2\,\delta_{ij}\right)$ is invertible, a set of vectors $\{\boldsymbol{\Delta}_n\}$ can be found such that the linear term vanishes. Thus

$$\mathcal{H}(\{\delta\tilde{\mathbf{R}}_l\}) = \sum_n \frac{\mathbf{P}_n^2}{2m} + \sum_n \frac{1}{2}m\omega^2\,\delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} U_{n,m}\,\delta\tilde{\mathbf{R}}_n\,\delta\tilde{\mathbf{R}}_m$$

At this point I introduce the quantized displacements in form oft the quadrature operators

$$\delta\tilde{\mathbf{R}}_n = \sqrt{\frac{\hbar}{2m\omega}}\,(a_n + a_n^\dagger)$$
$$\mathbf{P}_n = i\sqrt{\frac{\hbar m\omega}{2}}\,(a_n^\dagger - a_n)$$

With that the Hamiltonian gets its final form

$$\mathcal{H} = \sum_n \hbar\omega\, a_n^\dagger a_n + \frac{\hbar}{2m\omega}\sum_{n,m} U_{n,m}\left(a_n\, a_m + a_n^\dagger a_m + a_n\, a_m^\dagger + a_n^\dagger a_m^\dagger\right)$$

Bibliography

[1] E. Prodan and H. Schulz-Baldes, *Bulk and Boundary Invariants for Complex Topological Insulators: From K-Theory to Physics*, Mathematical Physics Studies (Springer International Publishing, Cham, 2016).

[2] V. Peano and H. Schulz-Baldes, "Topological edge states for disordered bosonic systems", Journal of Mathematical Physics **59**, 031901 (2018).

[3] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, "Topological insulators and superconductors: tenfold way and dimensional hierarchy", New Journal of Physics **12**, 065010 (2010).

[4] M. Maffei, A. Dauphin, F. Cardano, M. Lewenstein, and P. Massignan, "Topological characterization of chiral models through their long time dynamics", New Journal of Physics **20**, 013023 (2018).

[5] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig, "Classification of topological insulators and superconductors in three spatial dimensions", Physical Review B **78**, 195125 (2008).

[6] I. Mondragon-Shem, T. L. Hughes, J. Song, and E. Prodan, "Topological Criticality in the Chiral-Symmetric AIII Class at Strong Disorder", Physical Review Letters **113**, 046802 (2014).

[7] J. T. Wilson, S. Saskin, Y. Meng, S. Ma, R. Dilip, A. P. Burgers, and J. D. Thompson, "Trapping Alkaline Earth Rydberg Atoms Optical Tweezer Arrays", Physical Review Letters **128**, 033201 (2022).

[8] C. Hölzl, A. Götzelmann, E. Pultinevicius, M. Wirth, and F. Meinert, "Long-Lived Circular Rydberg Qubits of Alkaline-Earth Atoms in Optical Tweezers", Physical Review X **14**, 021024 (2024).